

Project Initialization and Planning Phase

Date	12 August 2026
Team ID	xxxxxxx
Project Title	India's agricultural crop production analysis (1997-2021)
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To Analyze India's agricultural crop production data from 1997 to 2021. It includes data cleaning, analysis and comparison of crop production across different years, crops and regions, followed by visualization through an interactive dashboard.
Scope	The project covers agricultural crop production in India from 1997 to 2021. It includes data cleaning, analysis and comparison of crop production across different years, crops and regions, followed by visualization through an interactive dashboard.
Problem Statement	
Description	Agricultural production data is large and difficult to analyze manually, Farmers, researchers and planners may find it difficult to identify crop production trends and compare production across different years, crops and regions.
Impact	Analyzing the data can help users understand historical crop production trends identify high-and low- production crops or regions, and support better agricultural planning and decision-making.
Proposed Solution	
Approach	Collect and clean agricultural crop production data from 1997-2021 perform exploratory data analysis, calculate relevant statistics and

	compare production across years, crops and regions. The analyses data will be presented using interactive charts and dashboards in Tableau for easy interpretation.
Key Features	• Years- wise crop production analysis • Crops- wise production comparison • State/ region-wise analysis • Identification of production trends • Interactive Tableaus charts and dashboard

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	2 x NVIDIA V100 GPUs
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy
Development Environment	IDE, version control	Jupyter Notebook, Git
Data		
Data	Source, size, format	Kaggle dataset, 10,000 images